

ARCHIMEDES Data Submission

1. Folder Structure

All data for your project must be placed inside your project's designated folder.

```
YourProjectName/
├── deidentified-raw/
│   ├── clinical/
│   ├── bids/
│   └── imaging/
│       ├── dicoms/
│       └── png/
└── documentation/
    └── data_dictionary/
```

← YOUR DATA
← Behavioural / REDCap CSV
← BIDS imaging & EEG
← DICOM studies
← PNG Images
← REDCap data dictionary

2. Behavioural Data (REDCap)

2.1 Folder

```
YourProjectName/
├── deidentified-raw/
│   ├── clinical/
│   │   └── your_data_export.csv
└── documentation/
    ├── data_dictionary/
    │   └── your_data_dictionary.csv
```

← participant redcap data
← REDCap data dictionary export

2.2 Preparation

The following must be set up in the ARCHIMEDES platform before your data can be ingested. All of these are created by the ARCHIMEDES team as part of your project set up(Ideally done via GUI project creation). Ideal to confirm they exist and that your data file values match exactly.

Item	Example	Notes
project	CNMDP	Must match exactly existing in platform for the corresponding project
site	UOHI, The Royal/IMHR	Must match exactly existing in platform for the corresponding project
cohort	Affected, Control, Not Applicable etc	Must match exactly existing in platform for the corresponding project
redcap_event_name	visit_1_arm_1	Each REDCap event must be registered as a visit in the platform
dob	YYYY-MM-DD	Data of birth to register participant in the platform

2.3 Data File Template

Start from your standard REDCap data export. Before submitting, add the ARCHIMEDES data platform fields described below. Place your file as

deidentified-raw/clinical/your_redcap_data_file.csv

Green = standard REDCap export fields (usually already present). Dark blue = required for participant creation in the platform (dob and sex) -must be added before submission. Orange = preferred but can be left empty if not known (project, site, cohort) ; the platform will use PI defaults where possible. Medium blue = instrument data columns. Note: redcap_event_name maps to the visit_label registered in the platform.

study_id	redcap_event_name	project	site	cohort	dob	sex	...	demo_age	gad7_q1	gad7_q2	...more
P001	visit_1_arm_1	CNMDP	UOHI	Affected	1985-03	Female	...	39	2	1	...

2.4 Data Dictionary

Example:

<https://www.utsouthwestern.edu/about-us/administrative-offices/information-resources/ras/assets/redcap-database-creating-dictionary.pdf>. Place in documentation/data_dictionary/. Field names in the DD are expected to match the column names in the data file.

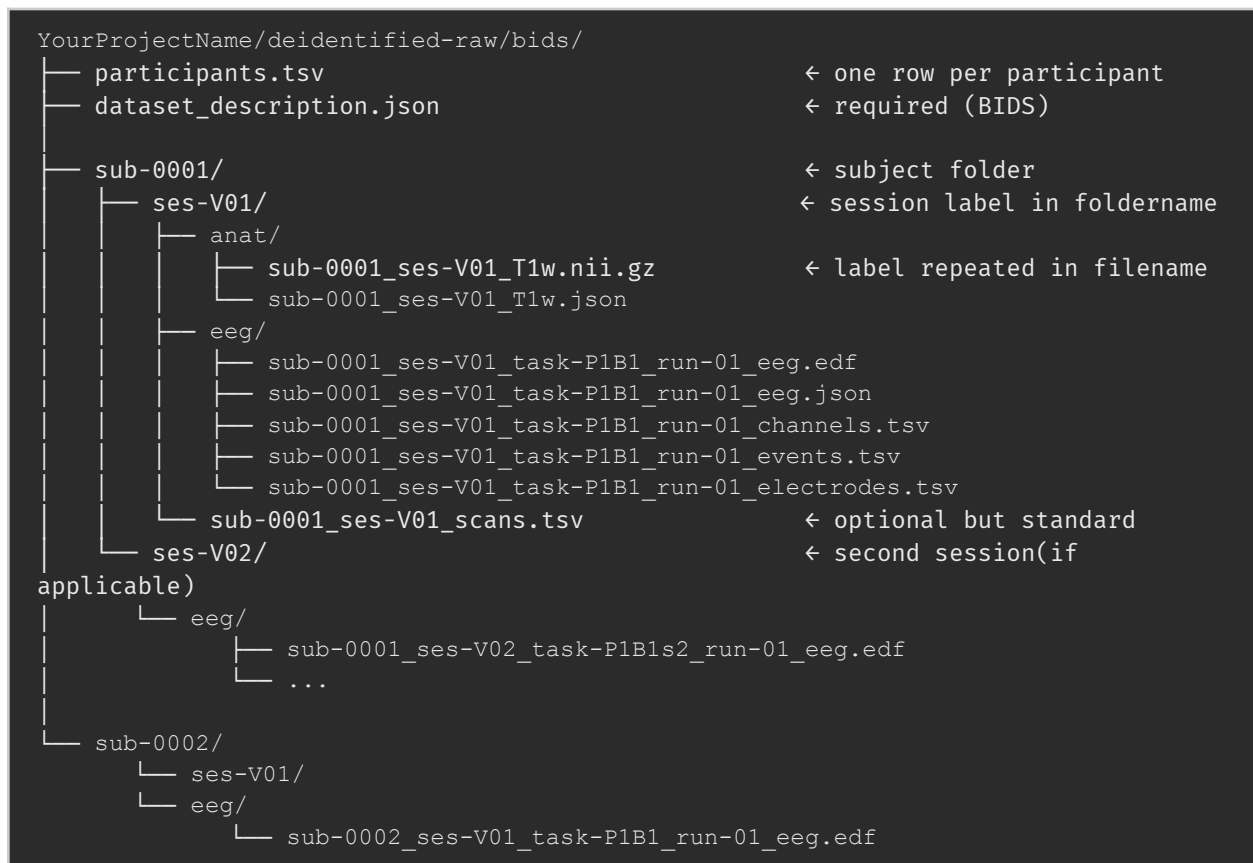
Longitudinal studies: the same participant appears on multiple rows; one row per visit/redcap_event_name.. The study_id stays the same (e.g. P001); only redcap_event_name changes.

3. BIDS Data (MRI & EEG)

MRI and EEG data must be in BIDS format. Both modalities share the same deidentified-raw/bids/ folder and the same participants.tsv.. For BIDS format examples:

<https://github.com/bids-standard/bids-examples>

3.1 Folder Structure



3.2 participants.tsv

One row per participant. participant_id must match the subject folder name exactly (e.g. sub-0001 corresponds to the sub-0001/ folder). The bare identifier (0001) is the same as study_id in the behavioural

data file. dob and sex are required for participant registration. project, site, and cohort are preferred but can be left empty if not known.

participant_id	sex	dob	site	cohort	project
sub-0001	female	1975-01	UOHI	Affected	CNMDP
sub-0002	male	1975-01	UOHI	Affected	CNMDP
sub-0003	female	1975-01	UOHI	Affected	CNMDP

4. DICOM Data

Place DICOM files in sub-folders under `deidentified-raw/imaging/dicoms/`. Each folder can contain DICOM files from multiple patients as exported from the scanner

4.1 Folder

```
YourProjectName/deidentified-raw/imaging/dicoms/  
├── StudyExport_2024-01/  
│   ├── IM-PatientA-0001.dcm  
│   ├── IM-PatientA-0002.dcm  
│   ├── IM-PatientB-0001.dcm  
│   └── IM-PatientB-0002.dcm  
└── StudyExport_2024-02/  
    ├── IM-PatientC-0001.dcm  
    └── IM-PatientD-0001.dcm
```

5. Date of Death Data Support

Date of Death (**dod**) is optional participant metadata. It can be supplied in one of two ways: in the same REDCap data file as the rest of the participant data, or in a separate registry file under the **registry/** directory, linked back to participants by **participant_id**. The pipeline scans both locations, so **dod** can be picked up from wherever it is maintained.

```
YourProjectName/  
├── deidentified-raw/ ← YOUR DATA  
│   ├── clinical/ ← contains REDCap data CSV data file with DoD header  
│   └── registry/ ← registry CSV - participant_id + dod columns
```

participant_id	dod
0001	1975-01-01
0002	1975-01-01
0003	1975-01-01